

Clustering

Prerequisites (Run on Both Machines)

1. Update System & Install MPI + SSH

```
sudo apt update  
sudo apt install -y mpich openssh-server build-essential
```

2. Add Host Entries for Name Resolution

Edit `/etc/hosts` on **both master and slave**:

```
sudo nano /etc/hosts
```

Add this (adjust IPs and hostnames as needed):

```
192.168.18.125 master iqra  
192.168.18.116 slave maryam
```

3. Verify SSH on Slave

On the **slave**:

```
sudo systemctl start ssh  
sudo systemctl enable ssh  
sudo ss -tlnp | grep :22 # Should show SSH is listening  
sudo ufw disable        # (Temporarily disable firewall)
```

Check from master:

```
ping slave # or use the IP
```

```
ssh slave # should ask for password
```

Create the **mpi** User (On Both Machines)

Run these on **both master and slave**:

```
sudo adduser --home /mirror --uid 1100 mpi
```

Then switch to **mpi** user:

```
su - mpi
```

Now **everything below should be done as **mpi** user**, unless stated otherwise.

Setup Passwordless SSH (As **mpi** user on Master)

1. Generate SSH Key

```
ssh-keygen -t rsa -N "" -f ~/.ssh/id_rsa
```

2. Copy SSH Key to Slave

```
ssh-copy-id mpi@192.168.18.116
```

If it fails with a permission error:

```
# First, remove old SSH fingerprints
ssh-keygen -R 192.168.18.116
# Then retry
ssh-copy-id mpi@192.168.18.116
```

Test it:

```
ssh mpi@192.168.18.116
# It should log in without asking password
```

Setup Shared Directory via NFS

On Master (Still as normal user):

```
sudo apt install -y nfs-kernel-server
sudo mkdir /mirror
sudo chown mpi:mpi /mirror
echo "/mirror *(rw, sync, no_subtree_check)" | sudo tee -a /etc/exports
sudo exportfs -a
sudo service nfs-kernel-server restart
```

On Slave:

```
sudo apt install -y nfs-common
sudo mkdir /mirror
sudo mount master:/mirror /mirror
```

Create and Compile MPI Program

On Master (as `mpi` user):

```
cd /mirror
gedit test.c
```

Paste this:

```
#include <mpi.h>
#include <stdio.h>
int main(int argc, char** argv) {
```

```

MPI_Init(&argc, &argv);
int rank, size;
MPI_Comm_rank(MPI_COMM_WORLD, &rank);
MPI_Comm_size(MPI_COMM_WORLD, &size);
printf("Hello from rank %d of %d\n", rank, size);
MPI_Finalize();
return 0;
}

```

Compile:

```
mpicc /mirror/test.c -o /mirror/test
```

Run the MPI Program

Still as `mpi` user on **master**, run:

```
mpiexec -n 2 -host 192.168.18.125,192.168.18.116 /mirror/test
```

If everything is configured correctly, you'll see:

```

Hello from rank 0 of 2
Hello from rank 1 of 2

```

Summary of `mpi` user shift

Step	As Which User?
System updates & installing MPI	<code>sudo</code> user
Editing <code>/etc/hosts</code>	<code>sudo</code> user
Creating <code>/mirror</code> , NFS setup	<code>sudo</code> user
Creating <code>mpi</code> user	<code>sudo</code> user
Everything after <code>su - mpi</code>	<code>mpi</code> user
Compiling & running code	<code>mpi</code> user